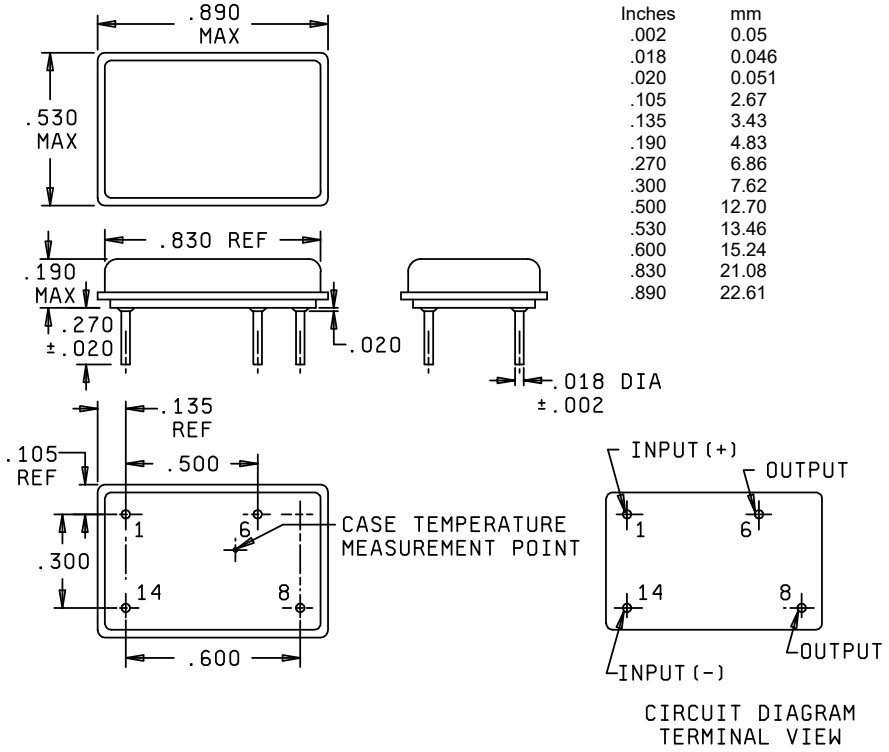
INCH-POUND
MIL-PRF-28750/8D
10 June 2021
SUPERSEDING
MIL-PRF-28750/8C
1 February 1999

PERFORMANCE SPECIFICATION SHEET

RELAY, SOLID-STATE, HERMETICALLY SEALED, CLASS II, OPTICALLY ISOLATED, 600 mA, 50 V dc, SIGNAL SWITCHING, SPST(N.O.)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and MIL-PRF-28750.



Inches	mm
.002	0.05
.018	0.046
.020	0.051
.105	2.67
.135	3.43
.190	4.83
.270	6.86
.300	7.62
.500	12.70
.530	13.46
.600	15.24
.830	21.08
.890	22.61

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal numbers shown above are for reference only.

FIGURE 1. Outline drawing and dimensions.

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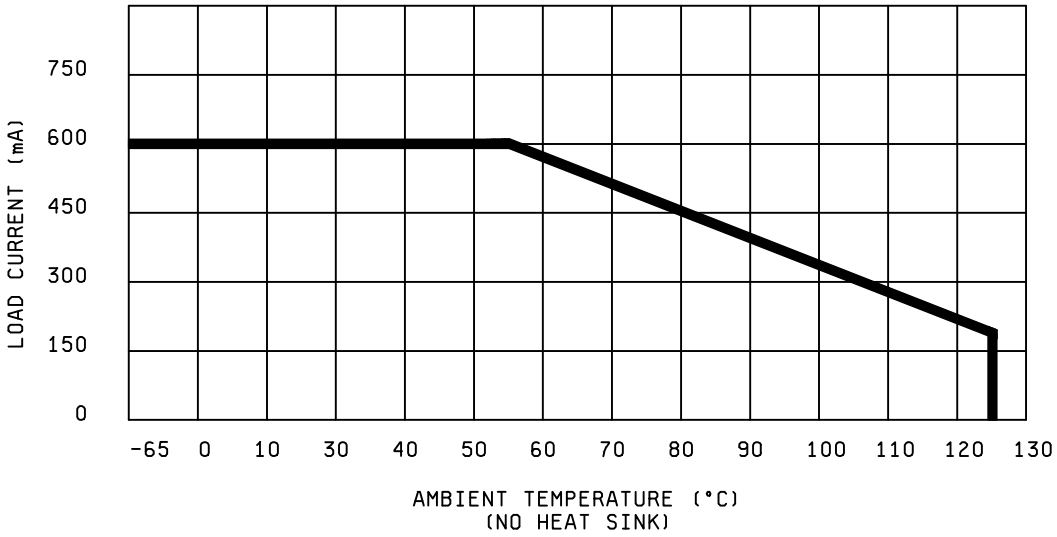


FIGURE 2. Maximum load current vs. ambient temperature.

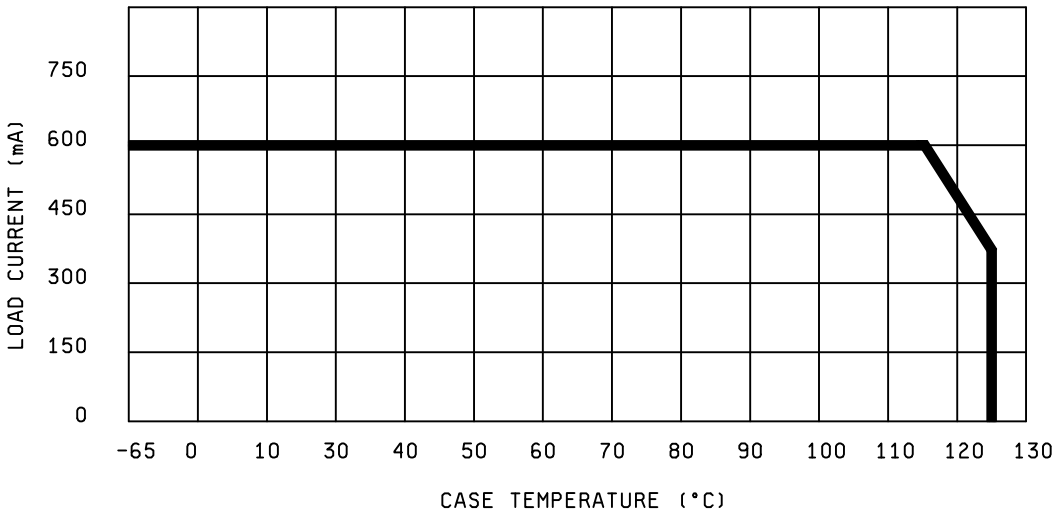


FIGURE 3. Maximum load current vs. case temperature.

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REQUIREMENTS:

MECHANICAL REQUIREMENTS:

Weight: 6 grams (0.21 ounce) maximum.

Seal: Hermetic.

Dimensions and configurations: See figure 1.

Terminals:

Terminal strength: 1 pound pull.

Terminal solderability: Applicable.

Terminal Finish: Corrosion resistant material. Corrosion resistant material such as gold plating are considered acceptable.

INPUT REQUIREMENTS:

Input voltage range: 3.0 V dc to 16.0 V dc.

Assured turn-on voltage: 3.0 V dc minimum.

Assured turn-off voltage: 1.0 V dc maximum.

Input current: 18 mA dc maximum.

Turn-on time: 50 microseconds maximum.

Turn-off time: 150 microseconds maximum.

Bias voltage: Not applicable.

OUTPUT REQUIREMENTS: (At +25°C unless otherwise specified.)

Rated output current: See figure 2 and figure 3.

Rated output voltage: 50 V dc maximum.

Output voltage drop: 1.4 V dc maximum.

Output leakage current: 60 microamperes maximum.

Minimum current: Not applicable.

Crosstalk: Not applicable.

Transient voltage: Not applicable.

Electric system spike: Not applicable.

Overload: Not applicable.

DC offset voltage: Not applicable.

Waveform distortion: Not applicable.

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ELECTRICAL REQUIREMENTS:

Dielectric withstanding voltage: 1,000 V ac rms, 60 Hz.

Insulation resistance: 100 megohms minimum.

Isolation: 5 picofarads maximum.

Power dissipation: 1.14 watts maximum.

Exponential rate of voltage rise (dv/dt): Not applicable.

ENVIRONMENTAL REQUIREMENTS:

Temperature:

Operating: -65°C to +125°C.

Storage: -65°C to +155°C.

Shock (specified pulse): MIL-STD-202-213, test condition G (100 g's).

Moisture resistance: Not applicable.

Vibration: MIL-STD-883, method 2007, test condition B (50 g's) except frequency shall be 10 Hz to 3,000 Hz.

Acceleration: 5,000 g's, Y1 axis, MIL-STD-883, method 2001, test condition A.

Resistance to soldering heat: Applicable.

Salt spray (corrosion): In accordance with MIL-STD-750, method 1041.

Marking: Marking shall include, as a minimum, the following information:

Part number.

Functional diagram and terminal identification.

Date code.

Manufacturer's name or source code.

The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-PRF-28750, this document references the following:

MIL-STD-202-213 MIL-STD-750 MIL-STD-883

Custodians:
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2021-011)

Review activities:
Navy - OS
Air Force – 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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